How small a state?
How simple a predictor?

The question

Can an image-based world model learn a code near its environment's state dimension, and can a small predictor evolve that code accurately? The main comparison is latent size × predictor capacity.

We built a controlled environment

Two colored balls in a unit 3D box with wall and ball collisions. Positions and velocities give 2 × balls × dimensions = 12 physical-state coordinates. The input is three RGB frames with XY/XZ/YZ views. State size and transition complexity are distinct: collisions introduce conditional dynamics.

We varied two capacities

Fixed 12-block, width-128 image encoder; latent sizes 6 / 12 / 24 / 48. Each encoder learns for 20k updates with a common MLP-256 predictor. Freeze it, then fit affine / MLP-64 / MLP-256 transitions for 10k updates each. Encode the initial history once; roll out 32 steps (1.28 s) without future images.

The completed primary sweep

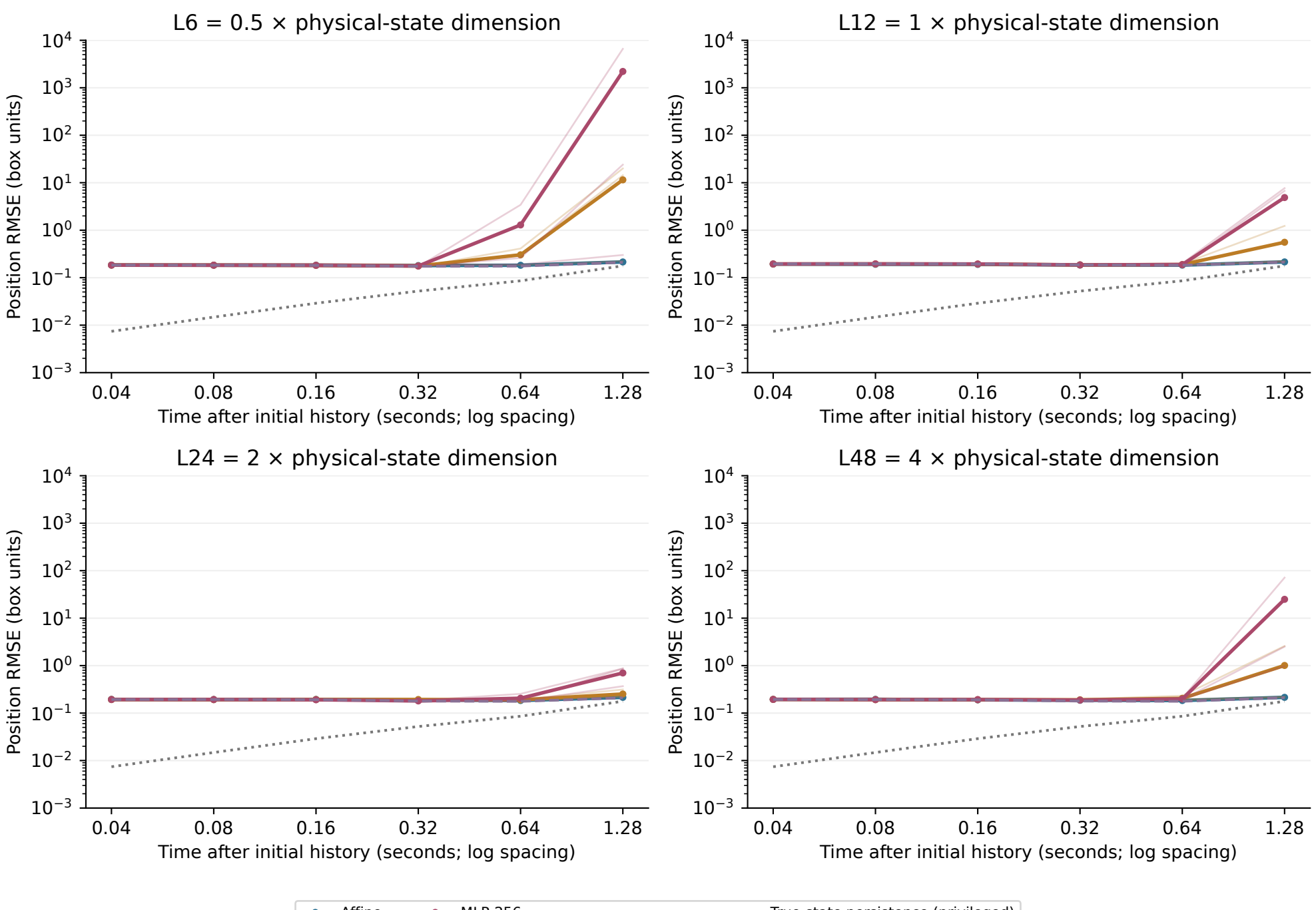
All four Gaussian latent sizes × three predictors × three seeds: 12 encoders and 36 frozen-predictor fits, tested on 256 episodes. Representation-family comparisons are secondary. The following pages show time versus physical RMSE for every setting, including all seeds.

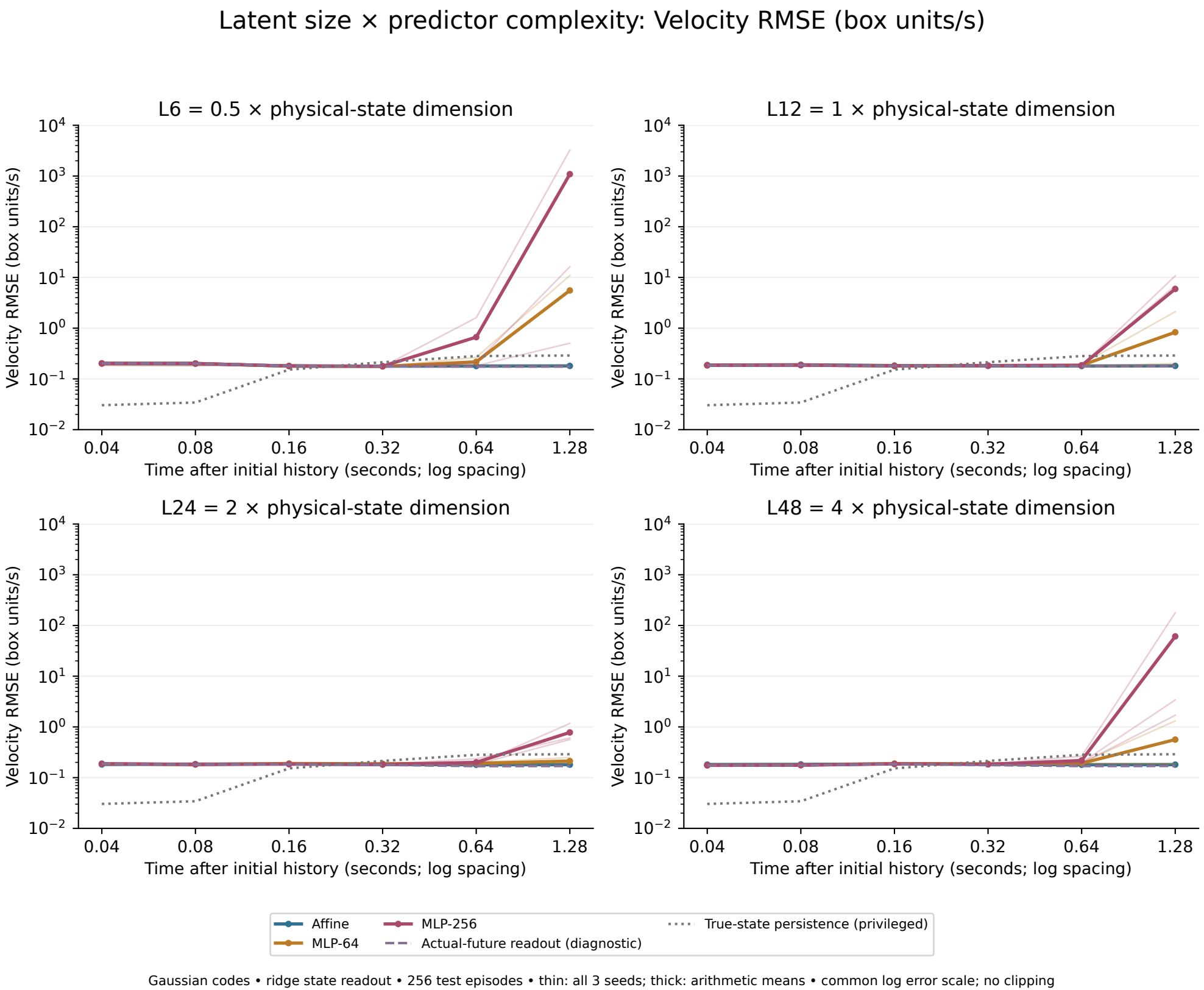
Long-horizon position results

The table below uses the same ridge state readout. Low error is better; large finite errors remain included. A small code has not automatically made prediction easier.

Latent	L / 12	Affine	MLP-64	MLP-256
6	0.5×	0.2144	11.5064	2208.7420
12	1×	0.2147	0.5574	4.8777
24	2×	0.2145	0.2523	0.7008
48	4×	0.2146	1.0078	24.9261

H32 position RMSE, box units; ridge readout; arithmetic means of 3 seeds.


Latent size × predictor complexity: Position RMSE (box units)
L6 = 0.5 × physical-state dimension
L12 = 1 × physical-state dimension
L24 = 2 × physical-state dimension
L48 = 4 × physical-state dimension
Position RMSE (box units)
Time after initial history (seconds; log spacing)
0.04
0.08
0.16
0.32
0.64
1.28
Affine
MLP-64
MLP-256
Actual-future readout (diagnostic)
True-state persistence (privileged)
Gaussian codes • ridge state readout • 256 test episodes • thin: all 3 seeds; thick: arithmetic means • common log error scale; no clipping


Latent size × predictor complexity: Velocity RMSE (box units/s)
L6 = 0.5 × physical-state dimension
L12 = 1 × physical-state dimension
L24 = 2 × physical-state dimension
L48 = 4 × physical-state dimension
Velocity RMSE (box units/s)
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
0.04
0.08
0.16
0.32
0.64
1.28
Time after initial history (seconds; log spacing)
Affine
MLP-64
MLP-256
Actual-future readout (diagnostic)
True-state persistence (privileged)
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What the sweep says about the hypothesis

Latent size and predictor complexity interact

At L6, nonlinear long-horizon errors are much larger than at L12 or L24; increasing from L24 to L48 does not consistently help. Affine error stays near 0.214 across sizes. There is no monotonic rule here that a smaller or larger latent makes a simpler predictor sufficient.

Local fit is not stable forecasting

At L12, native one-step MSE falls from 0.139842 (affine) to 0.001865 (MLP-64) and 0.001510 (MLP-256). MLP-256 H32 position errors are 0.218 / 6.736 / 7.679 across seeds. Matching physical-state dimension plus increasing predictor capacity is insufficient by itself.

State recovery and controls qualify the result

Actual-future-image ridge readout error is already about 0.21, so similar forecast errors do not establish useful compression. True-state persistence has stronger inputs (0.1777 position, 0.2892 velocity). Affine visual velocity is about 0.181. Learned true-state affine / MLP-256 position errors are about 0.1759 / 0.1756; encoder dimension alone is not isolated as the limiting factor.

Completed and already queued

Complete: the primary Gaussian sweep, Sparse L6 and nine true-state controls (15 encoders / 45 predictors / 30 readouts total). Queued: Sparse L12/24/48, their predictors/readouts, three supervised image-state references and final validation. The references test visual state recovery, not forecasting.

The environment-complexity question needs a follow-up

Vary ball count, spatial dimension and collision regimes; repeat matched latent/predictor sweeps around each 2nd state size. This is proposed, not queued. Current results come from one fixed environment and cannot establish a cross-environment capacity scaling law.

Visual aid and interpretation

Use the demo to connect time/RMSE curves to exact, predicted and actual-future readout trajectories. We have measured the tradeoff, but have not identified a minimum sufficient latent or necessary predictor width. Three-seed fixed-budget results do not claim convergence or equivalence.

Visual aid: https://henry550xz.github.io/jepa-billiards-demo/capacity.html
Scientific source: 968e6a825be49d662f303cf630315024f2f86d33
Numerical snapshot SHA256: 21fe079122d3decc68e57b6e51004abb439a26726aebedb5e6a59aba025b844a

Recorded results: 5 Nov 2026 | 256 test episodes | Environment / latent / predictor capacity